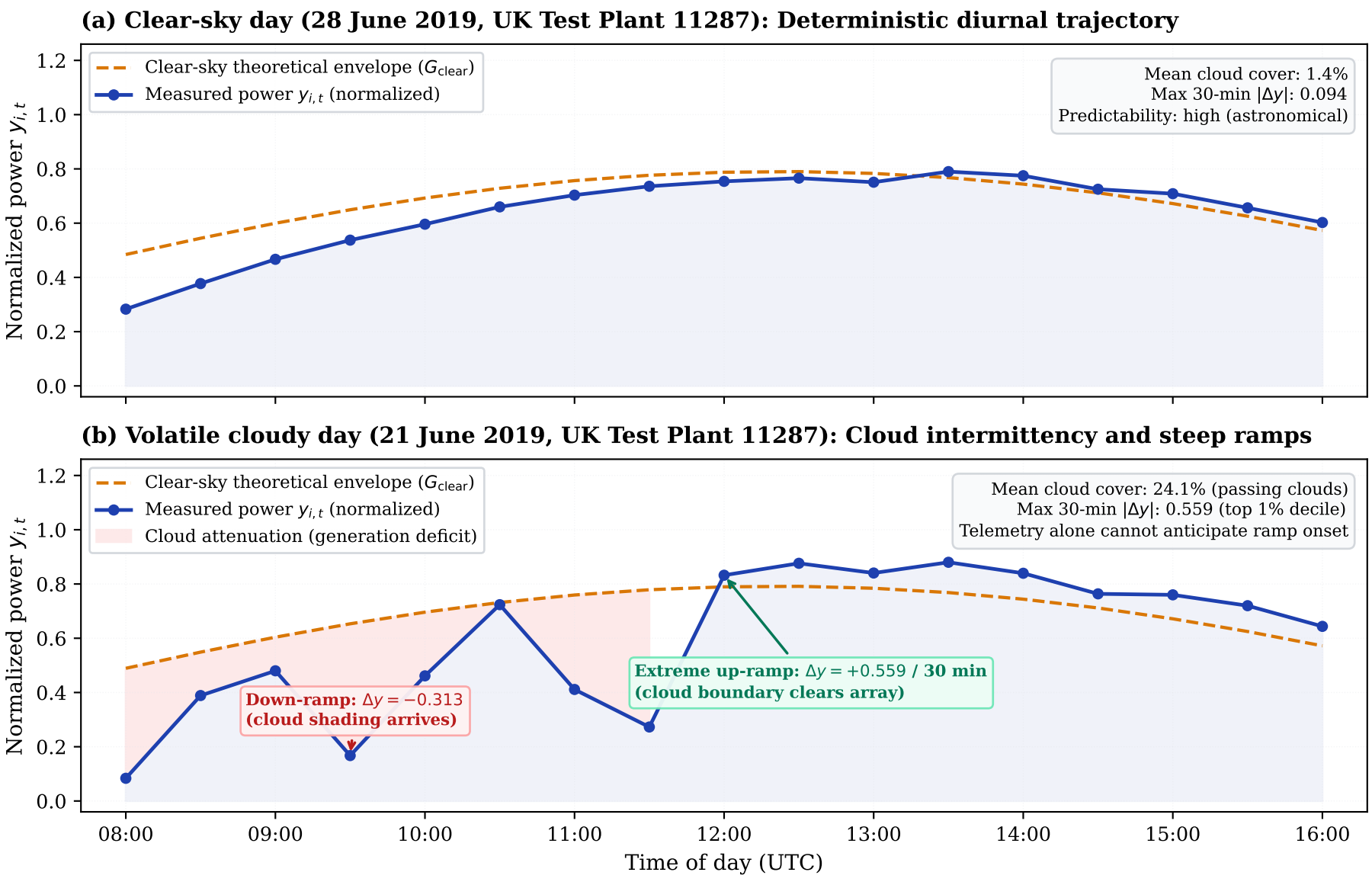
(a) Clear-sky day (28 June 2019, UK Test Plant 11287): Deterministic diurnal trajectory
Clear-sky theoretical envelope (G_{clear})
Measured power $y_{i,t}$ (normalized)
Mean cloud cover: 1.4%
Max 30-min |Δy|: 0.094
Predictability: high (astronomical)
Normalized power $y_{i,t}$
(b) Volatile cloudy day (21 June 2019, UK Test Plant 11287): Cloud intermittency and steep ramps
Clear-sky theoretical envelope (G_{clear})
Measured power $y_{i,t}$ (normalized)
Cloud attenuation (generation deficit)
Mean cloud cover: 24.1% (passing clouds)
Max 30-min |Δy|: 0.559 (top 1% decile)
Telemetry alone cannot anticipate ramp onset
Down-ramp: Δy = −0.313
(cloud shading arrives)
Extreme up-ramp: Δy = +0.559 / 30 min
(cloud boundary clears array)
Normalized power $y_{i,t}$
0.0
0.2
0.4
0.6
0.8
1.0
1.2
08:00
09:00
10:00
11:00
12:00
13:00
14:00
15:00
16:00
Time of day (UTC)